Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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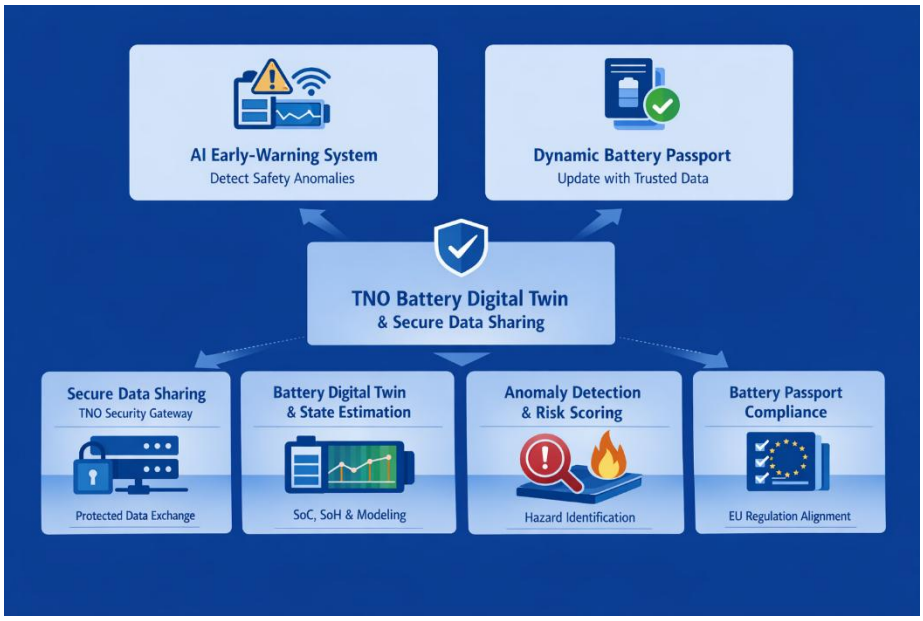
2. Technology Overview

Title of Proposed Technology	Secure, EU-compliant Dynamic Battery Passport and AI Early-Warning System for EV Battery Safety, Residual Value and Lifecycle Traceability			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity (√), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical/Electronics		Information/Communication	
	Medium Category	Small Category	Medium Category	Small Category
	Batteries	Secondary Batteries	Information Security	Service and Application Security
Concept of Proposed Technology	Core idea: combine TNO's validated battery modelling and test capabilities with sovereign data-sharing infrastructure to create: 1) an AI early-warning system for battery safety anomalies, and			

2) a Dynamic Battery Passport service layer that can be updated with trusted operational data.



Key functional blocks:

• Secure data acquisition and sharing: Use the TNO Security Gateway (TSG), an IDS (International Data Spaces) connector, to exchange BMS and operational data under enforceable data usage policies. [R3-R5]

• Battery state estimation and digital twin: Develop physics-based and hybrid (physics + ML) models for SoC, SoH, internal resistance growth and energy-based health metrics suitable for certification and valuation workflows. Build on TNO's battery modelling and BMS toolchain (incl. BMST). [R9-R11]

• Anomaly detection and risk scoring: Develop interpretable anomaly detection models (autoencoder-based, sequence models, change-point detection) to detect early indicators of unsafe behavior such as thermal runaway precursors, overcharge, cooling failure, and internal shorts. [R10-R11]

• Battery Passport alignment: Map data attributes and access rights to the EU Battery Regulation Battery Passport requirements (Article 77 and Annex XIII) and implement an interoperable data model and update mechanism. [R1-R2]

Potential use cases:

• EV fleet safety monitoring and early warning during charging and operation

• Residual value estimation for second-life decisions

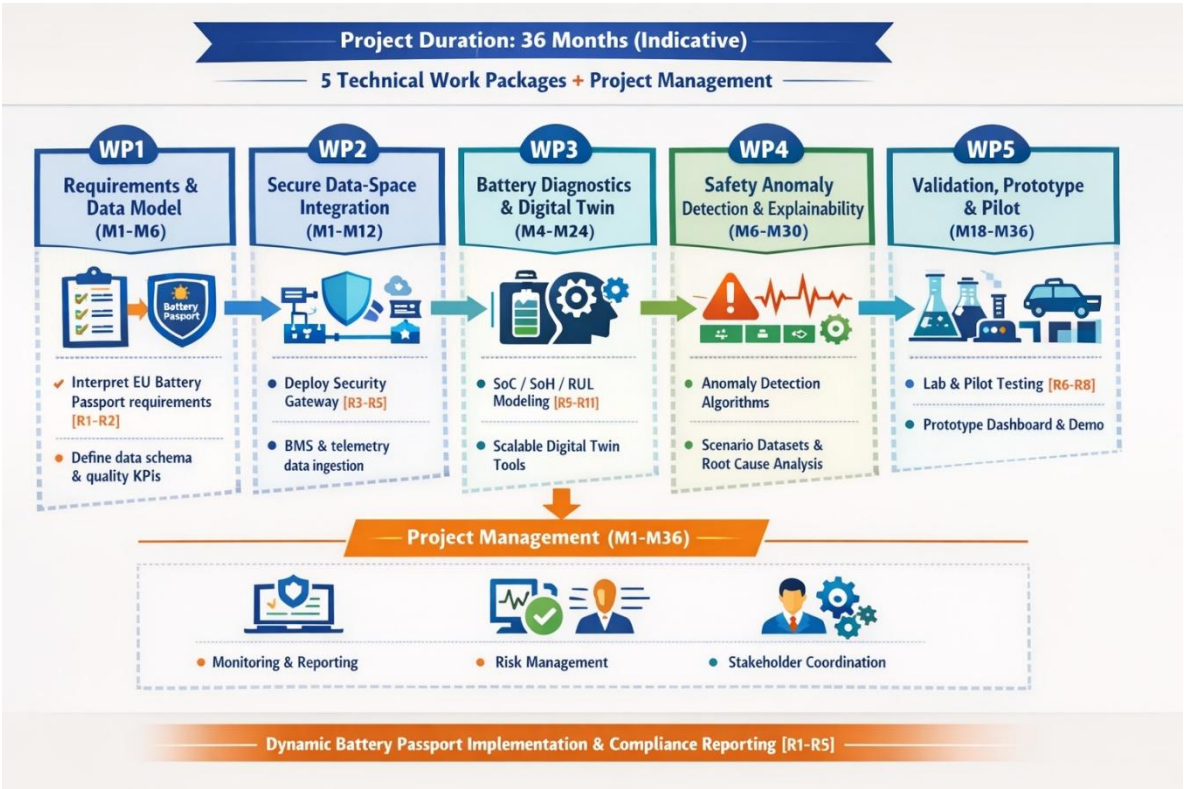
• Automated compliance data delivery for EU market entry (battery passport and reporting)

Rationale for Joint R&D Project	The EU Battery Regulation requires an electronic battery passport for EV batteries and certain industrial batteries, including static information and dynamic performance-related attributes. [R1-R2] Key bottlenecks for many value chain actors are: • Obtaining trusted operational data while preserving data sovereignty and business confidentiality • Producing traceable, explainable safety and health indicators beyond raw BMS estimates • Validating algorithms across realistic stress conditions and representative battery hardware This joint R&D combines: • TNO: independent validation, modelling and test infrastructure, and secure data space technology (TSG). [R3-R8] • Korean SME partner: real-world BMS and operational datasets, product integration, and route-to-market.
Key Words	1. Dynamic battery passport 2. Battery anomaly detection 3. Physics-informed battery diagnostics 4. International Data Spaces (IDS) 5. Battery digital twin

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2)]	Start of the Project	Level (3-4)	End of the Project	Level (6)
R&D Objectives	• Define EU battery passport data requirements, access rights, and update logic for dynamic operating data (aligned to Regulation (EU) 2023/1542). [R1-R2] • Implement an IDS/TSG-based secure data exchange between the Korean SME partner systems and TNO analytics services, including policy enforcement and audit logging. [R3-R5] • Develop and validate hybrid (physics + ML) state estimation and health diagnostics suitable for certification-style reporting (SoC, SoH, internal resistance, energy-based capacity metrics). [R9-R11] • Develop and validate interpretable early-warning anomaly detection and risk scoring for safety-critical events and precursors under diverse conditions. [R10-R11]			
R&D Target Specifications	• Data pipeline: IDS/TSG connector integration supporting policy-based access control; latency target < 10 s from data capture to analytics ingestion (prototype). [R3-R5] • Safety anomaly detection: time-to-warning target >= 5 min for defined precursor patterns; false positive rate target < 1 alert per vehicle per week in steady-state operation. • State estimation: energy-based capacity estimate error target <= 3% (kWh) versus laboratory reference; internal resistance trend error target <= 5% versus reference characterization. • Model portability: support at least 3 representative cell chemistries or pack configurations provided by the Korean SME partner, using parameter identification and transfer learning. • Validation coverage: tests across temperature range -45°C to +55°C in climatic-altitude environment and extended thermal envelope for pack testing where available. [R7-R8]			

R&D Plan



Project duration: 36 months (indicative), structured in 5 technical work packages plus project management.

WP1 Requirements and data model (M1-M6):

• Interpret EU battery passport requirements (Article 77 and Annex XIII) and derive a machine-readable data schema and access control matrix. [R1-R2]

• Define data quality KPIs, traceability rules, and validation protocols for dynamic data items.

WP2 Secure data-space integration (M1-M12):

• Deploy TNO Security Gateway (TSG) connectors and establish contract and policy templates for data sharing. [R3-R5]

• Implement ingestion adapters for BMS telemetry, charging logs, and contextual signals (as allowed by data policies).

WP3 Battery diagnostics and digital twin (M4-M24):

• Implement physics-based and hybrid diagnostics for SoC/SoH/internal resistance/RUL; include parameter identification and uncertainty quantification. [R9-R11]

• Integrate models in a scalable toolchain (offline training, edge-capable inference).

WP4 Safety anomaly detection and explainability (M6-M30):

• Develop interpretable anomaly detection (autoencoder, sequence models, change-point

	detection) and root-cause attribution. • Construct abnormal scenario datasets using laboratory stress testing and field events where available, including label governance and privacy controls. WP5 Validation, prototype and pilot (M18-M36): • Laboratory validation at TNO using battery cell, module and pack facilities (cycling, EIS, climatic chambers) and defined abuse and stress scenarios. [R6-R8] • Integrate algorithms into a prototype dashboard and APIs; conduct pilot with the Korean SME partner in a relevant environment. • Demonstrate a Dynamic Battery Passport update service with audit trail and compliance reporting export. [R1-R5]

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Provide secure data-sharing architecture and implementation guidance using IDS/TSG (connector deployment, policy templates, security review). [R3-R5] • Develop modelling and diagnostics algorithms (state estimation, degradation, anomaly detection), including explainability and uncertainty handling. [R9-R11] • Execute independent laboratory validation using TNO battery testing facilities and climatic chambers; generate reference datasets and validation reports. [R6-R8] • Lead EU compliance interpretation for battery passport data requirements and map project outputs to reporting needs. [R1-R2]
Expected R&R of Korean SME Partner	• Provide representative battery systems and BMS telemetry (data streams and documentation) and define the target deployment environment (EV fleet, charging, or ESS). • Support integration of the IDS/TSG connector into the SME's IT/OT environment and define data sharing policies and consent boundaries. • Contribute domain knowledge on failure modes, operational constraints, and service workflows; support evaluation of false alarms and usability. • Lead productization activities on the Korean side (embedded integration, UI/UX, customer pilots) and commercialization planning for EU and global markets.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	• Technical: validated prototype combining secure data exchange (IDS/TSG), hybrid battery diagnostics and interpretable anomaly detection, demonstrated on representative battery hardware and operating conditions. [R3-R8] • Compliance: reference implementation blueprint for EU-aligned Dynamic Battery Passport updates (data schema, access rights, audit trail, export interfaces). [R1-R2] • Commercial: pilot-ready software components (edge inference, cloud analytics, dashboard, APIs) enabling the Korean SME partner to offer battery safety and compliance services for EU market entry. • Scientific and dissemination: joint publications and conference contributions where allowed, building on Steven Wilkins-related work on battery modelling and BMS algorithms. [R9-R11]

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization